



Ordinary Differential Equations

NonLinear Models

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ENGR 213**

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Today's Lecture

- Nonlinear models
 - ✓ Population dynamics
 - ✓ Logistic equation
 - ✓ Chemical reactions
 - ✓ Leaking tank

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 2.8**

□ Population Dynamics

- Exponential growth: the model for exponential growth begins with the assumption that $dP/dt = kP$ for some $k > 0$. (P is population at time t).

- In this model, **relative (or specific) growth rate** defined by $\frac{dP}{dt}P$ is assumed to be a constant, k .

- True cases of exponential growth over long periods of time are hard to find, because limited resources will at some time exert restrictions on the growth of a population.



$$\frac{dP}{dt}P$$

is expected to decrease as P increases in size.

- The assumption that the rate at which a population grows (or declines) is dependent only on the number present can be stated as:

$$\frac{dP}{dt}P = f(P)$$

density-dependent hypothesis

□ Logistic Equation

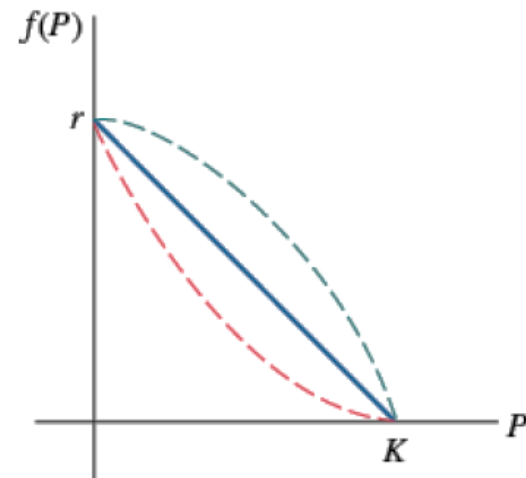
- If an environment is capable of sustaining no more than a fixed number of K individuals in its population.
- The quantity **K** is called the **carrying capacity** of the environment, and:

$$\frac{dP}{dt} = f(P) \longrightarrow f(K) = 0 \quad \text{and} \quad f(0) = r \quad (\text{initial condition})$$

- The simplest assumption that we can make is that $f(P)$ is linear:

$$f(P) = c_1 P + c_2 \xrightarrow[\text{conditions}]{\text{applying the}} \frac{dP}{dt} = P\left(r - \frac{r}{K}P\right)$$

logistic equation



$$\frac{dP}{dt} = P \left(\underbrace{r}_{\mathbf{a}} - \underbrace{\frac{r}{K}}_{\mathbf{b}} P \right) \longrightarrow \frac{dP}{dt} = P(a - bP)$$

$$\frac{dP}{dt} = P(a - bP) \quad \text{initial condition: } P(0) = P_0 \quad (\text{where } 0 < P_0 < a/b)$$

$$\frac{dP}{dt} = aP - bP^2$$

Inhibition or competition term

- Logistic functions (solutions of logistic equations) are quite accurate in predicting the growth patterns, in a limited space, of certain types of bacteria, protozoa, water fleas and fruit flies.

- Solution of the logistic equation:

$$\frac{dP}{dt} = P(a - bP)$$



EXAMPLE 1 (logistic growth)

Suppose a student carrying a flu virus returns to an isolated college campus of 1000 students. If it is assumed that the rate at which the virus spreads is proportional not only to the number x of infected students but also to the number of students not infected, determine the number of infected students after 6 days if it is further observed that after 4 days $x(4) = 50$.

□ Chemical reactions

- Suppose that chemicals A and B react and form chemical C.

e.g.,: $\text{CH}_3\text{Cl} + \text{NaOH} \rightarrow \text{CH}_3\text{OH} + \text{NaCl}$ (second-order reaction)

* for this reaction, the rate at which the reaction proceeds is proportional to the product of the concentrations of CH_3Cl and of NaOH .

- Let's say, **a** grams of A are combined with **b** grams of B. If there are M parts of A and N parts of B in C, and **X(t)** is the mass of C formed, then the mass of A and B remaining at any time are:

$$a - \frac{M}{N + M}X \quad \text{and} \quad b - \frac{N}{N + M}X$$

- The rate of the reaction will be:

$$\frac{dX}{dt} \propto \left(a - \frac{M}{N + M}X\right)\left(b - \frac{N}{N + M}X\right) \longrightarrow \frac{dX}{dt} = k(\alpha - X)(\beta - X)$$



EXAMPLE 2 (second-order chemical reaction)

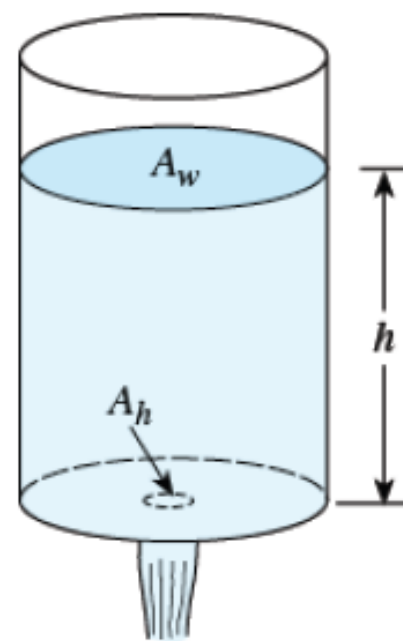
A compound C is formed when two chemicals A and B are combined. During the reaction, for each gram of A, 4 grams of B are used. It is observed that 30 grams of C are formed in 10 minutes. Initially there are 50 grams of A and 32 grams of B.

- (a) Determine the amount of C at time t if the rate of the reaction is proportional to the amounts of A and B remaining.
- (b) How much of the compound C is present at 15 minutes?
- (c) Interpret the solution as $t \rightarrow \infty$.

❑ Leaking Tank

- The speed v of efflux of water through a hole at the bottom of a tank filled to a depth h is: $v = \sqrt{2gh}$
- Imagine a water tank is draining through a hole with area of A_h , and we are interested in depth h of water remaining in the tank at time t .
- The volume of water leaving the tank per second is: $A_h\sqrt{2gh}$
- If $V(t)$ denotes the volume of water in the tank at time t : $\frac{dV}{dt} = -A_h\sqrt{2gh}$
- If the tank is such that volume of water in it at time t can be written $V(t) = A_w h$ (A_w is constant area of the upper surface of the water):

$$\frac{dV}{dt} = -A_h\sqrt{2gh} \xrightarrow{\frac{dV}{dt} = A_w \frac{dh}{dt}} \boxed{\frac{dh}{dt} = -\frac{A_h}{A_w}\sqrt{2gh}}$$





EXAMPLE 3 (leaking cylindrical tank)

A tank in the form of a right-circular cylinder standing on end is leaking water through a circular hole in its bottom.

(a) Solve for $h(t)$ if the initial height of water is H . Draw graph of $h(t)$ and give its interval I of definition in terms of the symbols A_w , A_h , and H .

(b) Suppose the tank is 10 ft high and has radius 2 ft and the circular hole has radius 0.5 in. If the tank is initially full, how long will it take to empty?

